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ASSIGNMENT # 02.

QUESTION NO. 05:

An air compressor takes in air at 100kPa, 17°C and delivers it at a 1MPa, 600K to a constant pressure cooler, which exits an air at 300K. Find the specific compressor work and specific heat transfer in the cooler.

Solution:

We are given the following data for Air compressor.

$T_1 = 17°C = 290K$

$T_2 = 600K$

$T_3 = 300K$

$P_1 = 100kPa$

$P_2 = 1MPa.$

Mass flow is given by:

$$\dot{m}_1 = \dot{m}_2 = \dot{m}_3$$

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Use Steady Flow Energy Equation:

$$h_1 + \frac{1}{2}V_1^2 + w = h_2 + \frac{1}{2}V_2^2$$

We have steady flow:

$$\boxed{V_1 \approx V_2 \approx 0}$$

Specific work is now equal to

$$w = h_2 - h_1$$

From Ideal Gas properties of Air Table A.7.1 corresponding to temperatures $T_1 = 290K$, $T_2 = 600K$. $T_3 = 300K$. we can obtain specific enthalpies:

At 290K:

$$h_1 = 290.43 \text{ KJ/Kg}$$

At 600K:

$$h_2 = 607.32 \text{ KJ/Kg}$$

At 300K:

$$h_3 = 300.47 \text{ KJ/Kg}$$

$$w = h_2 - h_1 = 607.32 - 290.43$$

$$\boxed{w = 316.9 \text{ KJ/Kg}}$$

Energy Equation

$$h_2 + \frac{1}{2}V_2^2 = h_3 + \frac{1}{2}V_3^3 + q$$

$$V_2 \approx V_3 = 0$$

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ific heat is equal to

$$q = h_2 - h_3 = 607.32 - 300.47$$

$$\boxed{q = 306.85 \text{ KJ/kg}}$$

QUESTION NO. 07: An irrigation pump takes water from a river at 10°C, 100 kPa and pumps it up to an open canal, where it flows out 1000 m higher at 10°C. The pipe diameter in and out of pump is 0.1 m and the power driving the unit is 5 hp. What is flow rate neglecting K.E & losses. ?

Solution: control volume around the pump and the piping up to the canal. This is a steady-flow process.

State 1 : (River):

$$P_1 = 100 \text{ kPa}$$

$$T_1 = 10°C \quad , \quad Z_1 = 0 \text{ m}$$

State 2 : (canal)

$$P_2 = 100 \text{ kPa (open to atmosphere)}$$

$$T_2 = 10°C,$$

$$Z_2 = 100 \text{ m}$$

Since the temperature and pressure are the same at both state, the specific enthalpies remains constant ($h_1 = h_2$)

Pipe diameter is constant 10.1mm at both inlet & outlet means

$(V_1 = V_2)$

$\Delta K.E = 0$

No heat Transfer $\Rightarrow (Q = 0)$

$1 hp = 0.745 kW$

$\dot{W}_{in} = 5hp \times 0.745 kW/hp$

$\dot{W}_{in} = 3.72 kW \Rightarrow 3728.5 \ J/s$

(Apply First law of Thermodynamics)

$$\dot{Q}_{in} + \dot{W}_{in} + \dot{m}\left(h_1 + \frac{V_1^2}{2} + gz_1\right) = \dot{Q}_{out} + \dot{W}_{out} + \dot{m}_2\left(h_2 + \frac{V_2^2}{2} + gz_2\right)$$

$Q = 0, \ h_1 = h_2, \ V_1 = V_2, \ z_1 = 0$

$\dot{W}_{in} = \dot{m} g z_2$

$\Rightarrow \dot{m} = \dfrac{\dot{W}_{in}}{g z_2}$

$\dot{m} = \dfrac{3728.5}{(9.81)(100)} = 3.80 \ kg/s$

QUESTION NO. 06 : A small expander (a turbine in heat transfer) has 0.05 kg/s helium entering at 1000 kPa, 550 K and it leaves at 250 kPa, 300 K. The power output on the shaft is measured to 55 kW. Find the rate of heat transfer neglecting kinetic energies.

SOLUTION : Helium is an ideal monoatomic gas constant specific heat

$$C_p = 5.1926 \text{ kJ/kg.K}$$

- Steady-flow control volume.

$$\dot{m} = 0.05 \text{ kg/s}$$

$$T_1 = 550 \text{ K}$$

$$T_2 = 300 \text{ K}$$

$$\dot{W} = 55 \text{ kW} \quad (55 \text{ kJ/s})$$

(First law of) conservation of Energy

$$\dot{Q} - \dot{W} = \dot{m}(h_2 - h_1)$$

For an ideal gas

$$h_2 - h_1 = C_p (T_2 - T_1)$$

$$\dot{Q} - \dot{W} = \dot{m} C_p (T_2 - T_1)$$

$$\dot{Q} - 55 = 0.05 \times 5.1926 (300 - 550)$$

$$\dot{Q} - 55 = 0.05 \times 5.1926 (-250)$$

$$\dot{Q} - 55 = -64.91$$

$$\dot{Q} = 55 - 64.91$$

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$\dot{Q} = -9.91\,kW$

(The system loses 9.91 kW of heat to the surroundings)

